ANDRÉS MORENO NAVARRO

Embedded Systems · PCB Design · Fusion 360 CAD

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PROFILE

Electrical and Electronic Engineering student at UWA (part-time, finishing 2029). I design and build my own embedded hardware: ESP32 firmware, custom PCBs in KiCad and 3D-printed enclosures in Fusion 360. Working in logistics, airport security and hospitality since 2020 has taught me to train new staff, stay calm under pressure and keep customers informed. Fluent in English and Spanish.

TECHNICAL SKILLS

Embedded: ESP32 / ESP32-S3 / ESP32-C3 and Arduino in C++; WiFi, BLE, ESP-NOW, MQTT, OTA updates; I²C, SPI, UART, I²S, PWM; servos, motor drivers, displays and sensors

PCB & electronics: KiCad (4-layer boards, JLCPCB Gerber/BOM/pick-and-place), EasyEDA, Fusion 360 Electronics; SMD hand soldering and reflow; fault-finding with a multimeter

CAD & fabrication: Fusion 360 part and enclosure design for FDM 3D printing (Creality K1C, Orca Slicer)

Software & tools: Python scripting, Git, Linux, Arduino IDE / arduino-cli; VHDL on an Intel DE10-Lite FPGA (Quartus, ModelSim)

ENGINEERING PROJECTS

RetroDS v2 — ESP32-S3 handheld game console *2026*

- Designed the schematic and 4-layer PCB in KiCad: ESP32-S3-WROOM-1U, ST7789 display, MAX98357A I²S amplifier, microSD, MCP73871 Li-ion charger with load sharing and USB-C.
- Assembled the JLCPCB board myself, then traced a hidden supply-to-ground short and a brown-out reset loop with a multimeter.
- Ported the open-source NESCAT emulator to the new board (display, I²S audio, SD, saves): correct NTSC speed, sound with volume control and save games, in a 3D-printed case.

ESP32-S3 development board *2026*

- First PCB I had manufactured: 4-layer KiCad design with native USB-C, TLV75801 LDO, ESD protection, WS2812B LED and ENS210 sensor; USB pair routed for 90 Ω (IPC-2141) and 0 DRC errors.
- Worked first time: programs over USB, and it is now the main controller in my biped robot.

Bip — two-legged robot *(in progress)* *2026 – present*

- ESP32-S3 with a PCA9685 driving 10 servos, MPU-6050 IMU and sonar; XIAO ESP32-S3 camera/microphone head and an ESP32-C3 running the face screens.
- Firmware updates over WiFi, a PC link for speech, voice replies and face tracking, and a MuJoCo simulation that runs the real walking code.

Wireless SO-101 robotic arm *2026*

- Built and calibrated an open-source SO-101 leader/follower pair (12 STS3215 servos) for LeRobot teleoperation.
- Replaced the USB link to the PC with two ESP32-C3 boards over ESP-NOW at 100 Hz with about 1–2 ms latency.

Other builds

- ESP-NOW remote-control cars with Fusion 360 chassis; a Telegram bot on an ESP32-C3 running 24/7 since June 2026; a BLE desk display with its own Android app; an Arduino exhaust-valve controller for a VW Golf.

Photos and write-ups: andresmoreno2702-gif.github.io · Build write-ups: hackster.io/andresmoreno2702

EDUCATION

Bachelor of Engineering (Honours) — Electrical & Electronic Engineering *Expected 2029*
University of Western Australia · Perth, WA · part-time, current

Diploma of Engineering *Graduated 01/2022*
Western Sydney University · Sydney, NSW · GPA 6.43 / 7 · Dean's Merit List

Bachelor of Engineering Science — Civil Engineering *(1.5 years, not completed)*
Western Sydney University · Sydney, NSW

WORK HISTORY

Logistics Coordinator | IKEA · Perth, WA *02/2024 – Current*

- Track customer orders and let customers know about their status or any delays.
- Train new co-workers on logistics procedures and working safely and efficiently.
- Work with customer service and other departments to resolve escalated orders.
- Coordinate the team, manage breaks and supervise shifts.

Airport Security Screener | Sydney Airport (Certis) · Sydney, NSW *02/2023 – 12/2023*

- Operated 2D and 3D X-ray machines for domestic and international cabin baggage.
- Carried out vehicle, frisk and baggage searches, explosive trace detection, walk-through metal detector and body scanner checks.
- Trusted with training new crew, machine calibrations, closing supervision and logging confiscated items.

Team Leader | Guzman y Gomez · Rouse Hill, NSW *12/2021 – 02/2023*

- Ran shifts, made sure procedures were followed and handled customer complaints.
- Main trainer for new staff; knew every station in the store, plus breaks and end-of-shift cash counts.

Crew Member | Guzman y Gomez · Parramatta, NSW *06/2020 – 01/2021*

- Worked every station, opened and closed the store, and regularly trained new crew.

AWARDS & CERTIFICATIONS

- **Dean's Merit List** — Diploma of Engineering 2021, Western Sydney University (awarded July 2022)
- **Udemy:** Fusion 360 Beginners Course · Complete PCB Design (EasyEDA) · Fusion 360 Circuit Design and PCB Manufacturing
- Certificate II in Transport & Security Protection · X-ray Operator (A-TIX & C-TIX) · Working With Children Check · Full driver licence

References available on request.